

AI Assistant Architecture & Starter Project

Great. Based on your choices:

Website Type: Full AI Assistant (chat, images, files, voice)

AI Model: Recommended option

Technology: HTML/CSS/JavaScript + Python

Hosting: Free hosting

Recommended Architecture

Frontend

- HTML
- CSS
- JavaScript

Backend

- Python + Flask

AI Services

- Text: Google Gemini API (good free tier)
- Images: Hugging Face APIs
- Voice: Browser Speech Recognition + Speech Synthesis

Database

- SQLite (free, simple)

Features

- AI Chat
- Voice Input
- Voice Output
- Image Generation
- File Upload (PDF, TXT)
- Chat History
- Mobile-Friendly UI
- Dark Mode

Project Structure

```
ai-assistant/  
  app.py  
  requirements.txt  
  database.db  
  static/  
  templates/  
  uploads/
```

Main Backend (app.py), Frontend (index.html), JavaScript, and deployment notes for Render, Railway, and others.